

DARKTRACE: Statistical Validation

1 Statistical Validation

All comparisons report central tendency with dispersion and significance, using McNemar’s test for paired classifier predictions, paired bootstrap for metric differences, exact Wilson score intervals for per-language accuracy, and a Friedman+Nemenyi omnibus test across ≥ 3 models, with Holm–Bonferroni correction over the resulting p -values ($\alpha = 0.05$). Tables ?? and ?? report measured values from the executed experiments (seed = 42); cells marked “re-run” require the prediction artifacts now emitted by the patched experiment scripts.

1.1 Classification and Severity Scoring

On the CoDA text corpus, both TF-IDF baselines are tightly estimated and the Linear-SVM advantage over Logistic Regression is statistically significant (McNemar $\chi^2 = 12.44$, $p = 4.19 \times 10^{-4}$; Table ??). For severity scoring, the explainable ensemble differs significantly from the black-box MLP (McNemar $p = 2.79 \times 10^{-4}$) while its 5-fold and bootstrap intervals overlap the baseline’s ranking metrics—i.e. DARKTRACE adds interpretability and better calibration without a statistically meaningful loss in discrimination.

Table 1: Classification and scoring significance (real). CIs are 95%; CV is 5-fold mean $\pm$ SD. McNemar is the paired test on the shared test set.

Task	Model	Macro-F1 [95% CI]	CV (mean $\pm$ SD)	Significance
Text (CoDA)	TF-IDF+LogReg	0.9034 [0.8896, 0.9161]	0.9129 $\pm$ 0.0082	McNemar $p = 4.19 \times 10^{-4}$
Text (CoDA)	TF-IDF+LinearSVM	0.9192 [0.9058, 0.9311]	0.9275$\pm$0.0075	
Severity	Explainable Ensemble	0.9290 [0.9187, 0.9394] [†]	—	McNemar $p = 2.79 \times 10^{-4}$
Severity	Black-box MLP	— (AUC gap +0.0045)	—	

[†] macro-F1 at the 0.5 decision threshold; bootstrap 95% CI.

1.2 Multilingual Detection: Per-Language Wilson Intervals

Table ?? reports exact Wilson 95% intervals for per-language accuracy. These make the small-sample caveat explicit and unavoidable: the apparently perfect Portuguese result (acc = 1.000) rests on only $N = 10$ items and carries a Wilson interval of [0.723, 1.000], and German/Spanish/French pooled cells ($N = 24/15/25$) are similarly wide. Only the well-sampled languages (EN $N \approx 1.7k$; RU $N = 114/542$) support firm claims. We therefore restrict headline multilingual conclusions to EN and RU and treat the remaining languages as exploratory pending more data.

1.3 Reproducing and Completing the Battery

The statistical layer is implemented in `src/metrics_stats.py` (McNemar, paired bootstrap, Wilson CI, Friedman+Nemenyi, Holm–Bonferroni) and driven by `src/exp_stats.py`, which

Table 2: Per-language accuracy with exact Wilson 95% CIs (real). “Reliable” flags $N \geq 30$. Macro-F1 per-language bootstrap CIs are added on re-run with saved predictions.

Regime	Lang	N	Macro-F1	Acc. [95% CI]	Reliable
Pooled	EN	1757	0.9078	0.899 [0.884, 0.912]	yes
	RU	114	0.9159	0.877 [0.804, 0.925]	yes
	FR	25	0.7439	0.960 [0.805, 0.993]	no
	DE	24	0.6528	0.917 [0.742, 0.977]	no
	ES	15	0.6787	0.800 [0.548, 0.929]	no
	PT	10	1.0000	1.000 [0.723, 1.000]	no
Transfer (EN→)	EN	1768	0.9160	0.908 [0.893, 0.920]	yes
	RU	542	0.4357	0.734 [0.696, 0.770]	yes
	DE	129	0.4425	0.876 [0.808, 0.922]	yes
	FR	99	0.5157	0.808 [0.720, 0.874]	yes
	ES	61	0.5727	0.836 [0.724, 0.908]	yes
	PT	54	0.6828	0.907 [0.801, 0.960]	yes

writes `stats_results.json` and `table_stats.csv`. The experiment scripts now persist per-instance predictions to `darktrace_results/preds/`; re-running `python -m src.exp_stats` after a real experiment run additionally emits (i) the paired-bootstrap macro-F1 difference for explainable-vs-black-box scoring, (ii) per-language macro-F1 bootstrap CIs, and (iii) the Friedman+Nemenyi critical-difference diagram across classifiers.